1:systemCompromised(f1tenth):0

2:RULE 63 (System compromised via drive DoS):0

3:drive_DoS(f1tenth):0

4:RULE 48 (Drive denial of service via topic flood):0

5:topicBelongsToSystem(’/drive’,f1tenth):1

41:RULE 49 (Unsafe command via gradual drift):0

6:controlTopic(’/drive’):1

8:RULE 16 (Attacker floods topic):0 43:RULE 71 (Attacker gradually drifts control values toward unsafe envelope):0

7:topicFlood(’/drive’):0

9:topicWrite(’/drive’):0

10:RULE 15 (Attacker writes topic):0

11:ddsSecurityDisabled(bridge):1

143:RULE 17 (Attacker calls service):0 73:RULE 14 (Attacker reads topic):0 157:RULE 14 (Attacker reads topic):0

12:rtpsReachableNode(bridge):0

93:RULE 12 (Attacker discovers ROS node through exposed DDS graph):0

13:RULE 0 (RTPS reachable node):0

14:netAccess(bridge,rtps,7400):1

15:attackerCanJoinRosGraph(robot_host):0

50:RULE 19 (Rogue publisher tampers sensor topic content):0

52:RULE 15 (Attacker writes topic):0

69:RULE 70 (Attacker runs MITM injection on topic):0

88:RULE 22 (Authorized rogue participant degrades sensor QoS):0

119:RULE 17 (Attacker calls service):0 133:RULE 17 (Attacker calls service):0

172:RULE 74 (Attacker creates new topic and dumps stolen data):0 170:RULE 74 (Attacker creates new topic and dumps stolen data):0

16:RULE 10 (Attacker can join ROS graph as an authorized participant):0

17:credentialAccess(robot_host):0

18:RULE 8 (Attacker obtains ROS credentials from reached host):0

20:lateralAccess(robot_host):0 19:hostStoresRosCredentials(robot_host):1

27:RULE 9 (Attacker is positioned on ROS network):0

101:RULE 11 (Attacker has host privilege for network shaping):0

21:RULE 7 (Attacker laterally reaches another host):0

22:hostReachable(engineering_ws,robot_host):1 23:initialAccess(engineering_ws):0

24:RULE 6 (Attacker gains initial access to an entry host):0

25:attackerEntryPoint(engineering_ws):1

26:attackerOnRosNetwork(robot_host):0

28:hostRunsRos(robot_host):1

29:nodeSubscribesToTopic(bridge,’/drive’):0

30:RULE 4 (Node subscribes to topic):0

31:typeOkSubscriber(bridge,’/drive’):0

32:RULE 2 (Subscriber type matches topic):0

34:topicType(’/drive’,’ackermann_msgs/msg/AckermannDriveStamped’):1 33:subType(bridge,’/drive’,’ackermann_msgs/msg/AckermannDriveStamped’):1

35:allowedSubscriber(’/drive’,bridge):1 36:topicSubscriber(’/drive’,bridge):1 37:rosTopic(’/drive’):1

38:topicAcceptsExternalPublish(’/drive’):1

39:RULE 64 (System compromised via unsafe command):0

40:command_unsafe(f1tenth):0

42:gradualDrift(’/drive’):0

44:RULE 65 (System compromised via scan corruption):0

45:scan_corrupted(f1tenth):0

46:RULE 50 (Scan corrupted via malicious content):0

47:topicBelongsToSystem(’/scan’,f1tenth):1

67:RULE 51 (Scan corrupted via MITM injection):0

86:RULE 53 (Scan unreliable via QoS degradation):0 81:RULE 52 (Scan corrupted via stale publish):0

94:RULE 54 (Scan unreliable via reliability flip):0

97:RULE 55 (Scan unreliable via topic drop):0 104:RULE 56 (Scan unreliable via topic delay):0 152:RULE 73 (Attacker correlates multiple sensitive topics):0

48:sensorTopic(’/scan’):1

96:RULE 75 (Attacker oscillates QoS reliability to disrupt subscribers):0

99:RULE 20 (Attacker drops sensor topic via network impairment):0 106:RULE 21 (Attacker delays sensor topic via network impairment):0

49:maliciousTopicContent(’/scan’):0

51:topicWrite(’/scan’):0

83:RULE 72 (Attacker publishes stale-timestamped messages):0 71:RULE 23 (Attacker can read and write topic):0

54:rtpsReachableNode(rviz):0 53:ddsSecurityDisabled(rviz):1

55:RULE 0 (RTPS reachable node):0

56:netAccess(rviz,rtps,7400):1

57:nodeSubscribesToTopic(rviz,’/scan’):0

58:RULE 4 (Node subscribes to topic):0

59:typeOkSubscriber(rviz,’/scan’):0

60:RULE 2 (Subscriber type matches topic):0

62:topicType(’/scan’,’sensor_msgs/msg/LaserScan’):1 61:subType(rviz,’/scan’,’sensor_msgs/msg/LaserScan’):1

77:RULE 1 (Publisher type matches topic):0

63:allowedSubscriber(’/scan’,rviz):1 64:topicSubscriber(’/scan’,rviz):1 65:rosTopic(’/scan’):1

75:RULE 3 (Node publishes to topic):0

66:topicAcceptsExternalPublish(’/scan’):1

68:mitmInjection(’/scan’):0

70:mitmPossible(’/scan’):0

72:topicRead(’/scan’):0

74:nodePublishesToTopic(bridge,’/scan’):0

76:typeOkPublisher(bridge,’/scan’):0

78:pubType(bridge,’/scan’,’sensor_msgs/msg/LaserScan’):1

79:allowedPublisher(’/scan’,bridge):1 80:topicPublisher(’/scan’,bridge):1

82:stalePublish(’/scan’):0

84:RULE 66 (System compromised via scan unreliability):0

85:scan_unreliable(f1tenth):0

87:qosDegradation(’/scan’):0

89:targetNodeSelected(bridge):0

90:RULE 13 (Attacker selects impactful target node):0

91:nodeAffectsSystem(bridge,f1tenth):1

141:RULE 60 (Bridge misconfigured via parameter tamper):0

92:graphDiscovery(bridge):0

95:reliabilityFlip(’/scan’):0

98:topicDrop(’/scan’):0

100:attackerCanNetAdmin(robot_host):0

102:hostHasNetAdmin(robot_host):1

103:nodeHostedOn(bridge,robot_host):1

105:topicDelay(’/scan’):0

107:RULE 67 (System compromised via map server down):0

108:map_server_down(f1tenth):0

109:RULE 57 (Map server down via lifecycle hijack):0

110:nodeAffectsSystem(map_server,f1tenth):1

138:RULE 59 (Map server down via direct parameter tamper):0 137:RULE 58 (Map server down via direct lifecycle abuse):0

111:lifecycleHijack(map_server):0

112:RULE 76 (Attacker hijacks lifecycle node with malicious config):0

113:serviceServer(’/map_server/change_state’,map_server):1

135:RULE 5 (Service exists on server):0

114:lifecycleService(’/map_server/change_state’):1

131:RULE 28 (Attacker abuses lifecycle service):0

115:parameterTamper(map_server):0

116:RULE 27 (Attacker tampers parameters):0

117:parameterService(’/map_server/set_parameters’):1

118:serviceCall(’/map_server/set_parameters’,map_server):0

120:ddsSecurityDisabled(map_server):1 121:rtpsReachableNode(map_server):0

122:RULE 0 (RTPS reachable node):0

123:netAccess(map_server,rtps,7400):1

124:rosService(’/map_server/set_parameters’,map_server):0

125:RULE 5 (Service exists on server):0

126:serviceType(’/map_server/set_parameters’,’rcl_interfaces/srv/SetParameters’):1 127:serviceServer(’/map_server/set_parameters’,map_server):1

128:lifecycleDisruption(map_server):0

129:RULE 37 (Lifecycle disrupted):0

130:lifecycleAbuse(map_server):0

132:serviceCall(’/map_server/change_state’,map_server):0

134:rosService(’/map_server/change_state’,map_server):0

136:serviceType(’/map_server/change_state’,’lifecycle_msgs/srv/ChangeState’):1

139:RULE 68 (System compromised via bridge misconfiguration):0

140:bridge_misconfigured(f1tenth):0

142:serviceCall(’/bridge/set_parameters’,bridge):0

144:rosService(’/bridge/set_parameters’,bridge):0

145:RULE 5 (Service exists on server):0

147:serviceServer(’/bridge/set_parameters’,bridge):1 146:serviceType(’/bridge/set_parameters’,’rcl_interfaces/srv/SetParameters’):1

148:RULE 69 (System compromised via data leak):0

149:data_leaked(f1tenth):0

150:RULE 61 (Data leaked via cross-topic correlation):0

151:crossTopicCorrelation(f1tenth):0

153:topicBelongsToSystem(’/odom’,f1tenth):1 155:sensitiveTopic(’/scan’):1 154:sensitiveTopic(’/odom’):1 156:topicRead(’/odom’):0

158:nodePublishesToTopic(bridge,’/odom’):0

159:RULE 3 (Node publishes to topic):0

160:typeOkPublisher(bridge,’/odom’):0

161:RULE 1 (Publisher type matches topic):0

163:topicType(’/odom’,’nav_msgs/msg/Odometry’):1 162:pubType(bridge,’/odom’,’nav_msgs/msg/Odometry’):1

164:allowedPublisher(’/odom’,bridge):1 165:topicPublisher(’/odom’,bridge):1 166:rosTopic(’/odom’):1

167:RULE 62 (Data leaked via attacker-created exfil topic):0

169:exfiltrateChannel(’/attacker_exfil’):0 168:rosTopic(’/attacker_exfil’):1

171:attackerCreatedTopic(’/attacker_exfil’):1